

Has the Probability that a Film is an Adaptation Increased After 2010?

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[melbrt/bayesian-analysis-film-adaptations](https://melbrt.github.io/bayesian-analysis-film-adaptations)

1. Introduction

The film industry has increasingly been associated with adaptations of pre-existing works such as novels, comic books, television series, video games, and theatrical productions. Contemporary audiences are often exposed to franchises and intellectual properties that already exist before being adapted into films.

This phenomenon has generated discussions among scholars, critics, and industry professionals. Some observers argue that modern cinema relies more heavily on existing intellectual properties than in previous decades, while others suggest that adaptation has always been an important component of film production.

The objective of this project is to investigate whether films released after 2010 are more likely to be adaptations than films released before 2010. To answer this question, a Bayesian statistical approach was applied to data collected from Wikidata.

2. Historical Context of Film Adaptations

Adaptation has been present since the earliest years of cinema. During the silent film era, filmmakers frequently adapted novels, myths, and theatrical works because these stories were already familiar to audiences.

Throughout the twentieth century, adaptation remained a common practice. Literary works often served as sources for film productions, allowing filmmakers to rely on narratives that had already demonstrated cultural success.

The development of blockbuster cinema during the late twentieth century contributed to the growing importance of adaptations. Major film studios increasingly invested in projects based on established intellectual properties because these productions were perceived as less risky than entirely original works.

The beginning of the twenty-first century saw the emergence of large-scale franchises such as Harry Potter, The Lord of the Rings, and the Marvel Cinematic Universe. These productions demonstrated the commercial value of adapting existing cultural products and contributed to the perception that modern cinema increasingly depends on pre-existing material.

The present project investigates whether this perception can be supported by empirical evidence.

3. Historical Background of Bayesian Statistics

Bayesian statistics originates from the work of Thomas Bayes, an eighteenth-century mathematician and Presbyterian minister. His essay, published posthumously in 1763, introduced a method for updating probabilities when new information becomes available.

The Bayesian framework was later expanded by Pierre-Simon Laplace, who generalized the approach and applied it to numerous scientific problems. Laplace demonstrated that probability could be interpreted as a measure of uncertainty and used Bayesian reasoning in astronomy, demography, and other quantitative disciplines.

Although Bayesian methods were known throughout the nineteenth century, they became less prominent during much of the twentieth century as frequentist statistics gained dominance in scientific research.

The development of modern computing dramatically changed this situation. Advances in numerical simulation and computational power made Bayesian methods increasingly practical. Today, Bayesian statistics is widely used in fields such as machine learning, artificial intelligence, medicine, economics, ecology, and social sciences.

Because Bayesian inference directly quantifies uncertainty and allows probabilistic comparisons between hypotheses, it provides an appropriate framework for addressing the research question examined in this project.

4. Why Use Bayesian Statistics?

The objective of this study is not only to estimate adaptation rates but also to compare them and evaluate the probability that one rate exceeds another. A Bayesian framework offers several advantages for this type of problem.

First, the variable under study is binary: a film is either an adaptation or it is not. This structure naturally corresponds to a Bernoulli process and can be modeled efficiently using a Beta-Bernoulli framework. Second, Bayesian inference provides complete posterior distributions rather than single point estimates. This allows uncertainty to be represented explicitly. Third, Bayesian methods allow direct computation of the probability that adaptation rates increased after 2010. Rather than relying on p-values, the analysis estimates the quantity of direct interest:

$$P(\theta_{\text{after}} > \theta_{\text{before}})$$

This probability directly answers the research question and facilitates interpretation of the results. For these reasons, Bayesian statistics provides a particularly suitable methodological framework for this project.

5. Visualisation of Results

Table 1 presents the number of films, the number of adaptations, and the observed adaptation rate in each period.

	Period	n (films)	k (adaptations)	Proportion
0	Before 2010	1492	417	0.279491
1	After 2010	2085	584	0.280096

Before 2010, 417 of 1492 films were adaptations (27.95%). After 2010, 584 of 2085 films were adaptations (28.01%). The observed proportions are nearly identical, suggesting little evidence of a substantial increase in adaptation rates after 2010. Bayesian inference is used in the following sections to account for statistical uncertainty and formally compare these probabilities.

5.1 Posterior Distributions

The posterior distributions of adaptation probabilities before and after 2010 were plotted on the same graph. These visualisations allow direct comparison of both probability distributions and facilitate interpretation of differences between periods. The density curves illustrate not only the estimated probabilities but also the uncertainty associated with these estimates. Figure 1 shows the posterior distributions of adaptation probabilities before and after 2010. The strong overlap between both distributions suggests very similar adaptation rates across periods.

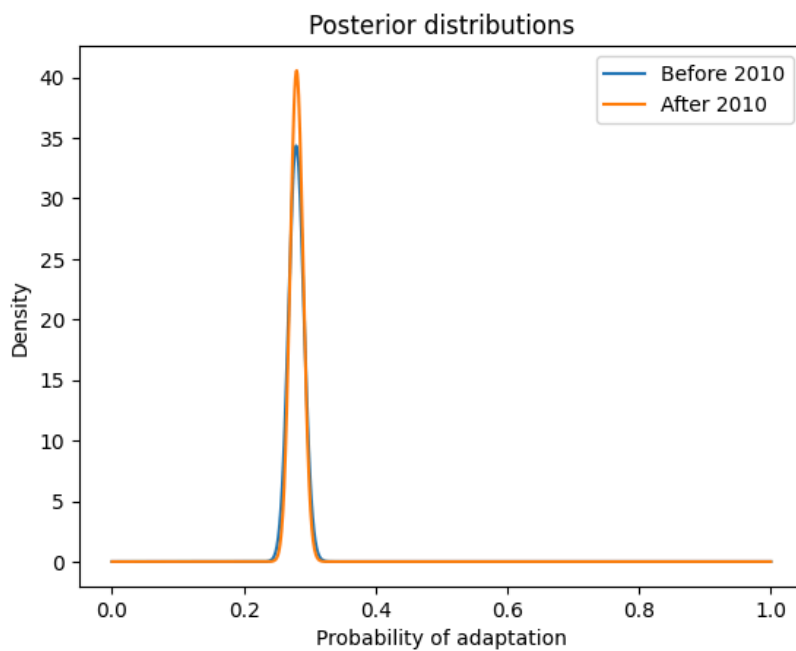


Figure 1. Posterior Distributions of Adaptation Probabilities

5.2 Posterior Distribution of the Difference

A second graph was produced to visualize the posterior distribution of:

$$\Delta = \theta_{\text{after}} - \theta_{\text{before}}$$

This representation provides a direct visual assessment of the evidence supporting an increase in adaptation rates after 2010. The vertical line at zero represents the absence of difference between periods. The position of the posterior distribution relative to this reference line allows straightforward interpretation of the results. Figure 2 displays this distribution, which is centered around zero, indicating little evidence for a substantial change in adaptation rates.

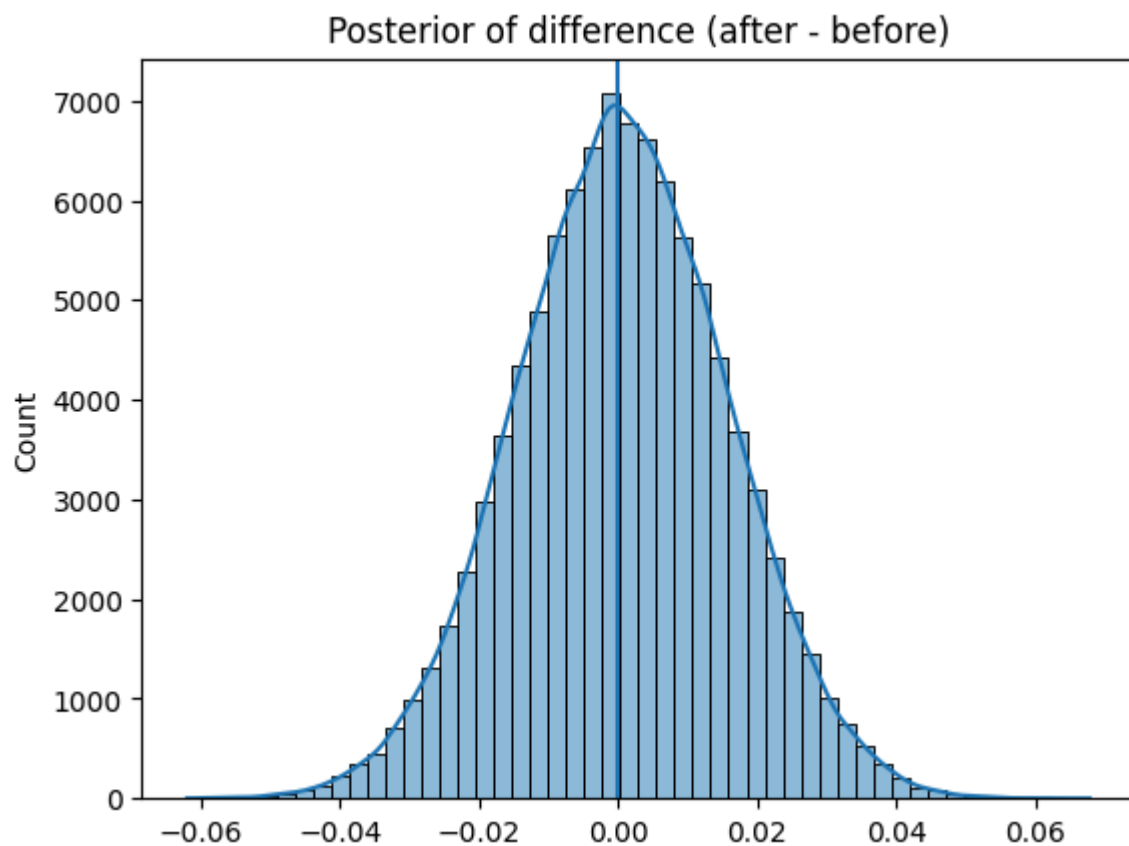


Figure 2. Posterior Distribution of the Difference (After – Before)

6. Results Interpretation

The descriptive statistics indicate very similar adaptation rates across the two periods. Before 2010, 417 out of 1,492 films were identified as adaptations, corresponding to an observed adaptation rate of 27.95%. After 2010, 584 out of 2,085 films were identified as adaptations, corresponding to an observed adaptation rate of 28.01%.

Bayesian inference leads to the same conclusion. The posterior mean adaptation probability is estimated at 0.280 for films released before 2010 and 0.280 for films released after 2010. The corresponding 95% credible intervals overlap almost completely:

- Before 2010: [0.257 ; 0.303]
- After 2010: [0.261 ; 0.300]

The posterior distributions displayed in Figure 1 are nearly identical, indicating that the estimated probabilities of adaptation are extremely similar across periods.

To directly compare the two probabilities, Monte Carlo sampling was performed. The posterior probability that adaptation rates increased after 2010 is:

$$P(\theta_{\text{after}} > \theta_{\text{before}}) = 0.512$$

This value is extremely close to 0.5, meaning that the posterior evidence is almost perfectly balanced between the two hypotheses. Furthermore, the 95% credible interval for the difference is [-0.0295 ; 0.0302]. Because this interval contains zero and remains narrow around it, the analysis provides no evidence for a meaningful change.

These results challenge the widespread perception that contemporary cinema relies substantially more on adaptations than earlier periods. While franchises and adaptations clearly play an important role in modern film production, the structured data analysed in this project do not indicate a statistically significant increase after 2010.

7. Limitations of the Study

Several limitations should be acknowledged. First, the analysis relies on information contained in Wikidata. Although it is a massive database, its completeness depends on community contributions; some adaptation relationships may be missing. Second, adaptation was measured using the strict binary Wikidata property "based on" (P144), which simplifies a complex cultural phenomenon. Third, the project compares only two broad periods. Finally, the analysis focuses exclusively on adaptation status and does not consider other variables such as budget, genre, or country of production.

8. Technical Conclusion

The objective of this project was to determine whether films released after 2010 are more likely to be adaptations than films released before 2010. Using a Beta-Bernoulli Bayesian model, posterior distributions were estimated. The results show that the adaptation rates are almost identical, with posterior means of approximately 28% in both groups. Consequently, the analysis does not provide sufficient Bayesian evidence to conclude that films released after 2010 are more likely to be adaptations. The results fail to support the alternative hypothesis H1 and are highly consistent with the null hypothesis H0.

Next steps: If this project had to be continued, several steps would be taken to deepen the analysis. First, rather than using a simple binary temporal split (before/after 2010), a longitudinal analysis grouping films by decade or year would be implemented to reveal more nuanced trends.

Second, a Bayesian logistic regression or a hierarchical model could be used to include other covariates (like budget or genre), as the "adaptation effect" might be heavily concentrated in high-budget blockbusters rather than independent films. Finally, cross-referencing Wikidata with IMDb or TMDB datasets would improve data robustness.

9. Epistemological Conclusion

From an epistemological perspective, this project illustrates the strengths and limitations of quantifying cultural phenomena. While Bayesian statistics provide a rigorous framework to move beyond anecdotal impressions, cultural concepts cannot be fully reduced to binary variables without losing nuance.

On a more personal note, the idea for this project emerged from a common cultural intuition: as a regular cinema-goer, I had the distinct impression that original ideas were disappearing in favour of remakes and adaptations. I wanted to confront this personal bias with hard data. To achieve this, the workload was organized sequentially: formulating the SPARQL query to extract reliable data, researching the mathematical background to select the Beta-Bernoulli model, writing the Python script, and finally interpreting the results.

If I were to realise this project again, I would definitely change how I collect the data. Relying strictly on the "based on" property proved a bit too restrictive. I would try to extract textual descriptions to capture "looser" forms of inspiration.

10. Bibliography

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